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Ministerie van Landbouw, Visserij, Voedselzekerheid en Natuur

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To the Minister of Health, Welfare and Sport and the Inspector-General of the Dutch Food and Consumer Product Safety Authority

Advice from the Director of the Office for Risk Assessment & Research

On the public health risks associated with offering cultured meat at tastings

Office for Risk Assessment & Research

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Background

Placing meat on the market made from in vitro cultured animal cells (cultured meat) is currently not formally authorised in Europe. Cultured meat is considered a novel food within the European Union (EU), whose market authorisation is only allowed after a safety assessment by the European Food Safety Authority (EFSA), followed by approval by the European Commission (EC) and the EU Member States.

In March 2022, a motion by members of parliament Tjeerd de Groot and Valstar (Parliamentary Paper 27 428, No 383) was adopted by the Dutch House of Representatives requesting the government to enter into consultations with Dutch producers of cultured meat in order to allow sensory evaluations, hereafter called 'tastings', under 'controlled and safe conditions'. Subsequently, in collaboration with the Ministry of Health, Welfare and Sport (VWS) and the Ministry of Agriculture, Fisheries, Food Security and Nature (LVVN), cultured meat producers and the Dutch association of biotechnology companies (HollandBio) have developed a Code of Practice (CoP) that prescribes the procedures to be followed when conducting cultured meat tastings. In April and July 2024, two Dutch producers of cultured meat organised tastings for the first time in the Netherlands to have their cultured meat products tasted by a number of pre-selected participants. Tastings also took place in February 2025.

The Dutch Food and Consumer Product Safety Authority (NVWA)-wide working group 'Room for experimentation with Cultured Meat' wants to form its own opinion on food safety during cultured meat tastings and has asked The Office for Risk Assessment & Research (BuRO) the following questions:

Question 1: *What are the chemical and (micro)biological risks associated with the tasting of cultured meat?*

Question 2: *What (scientific) knowledge has already been gained about this abroad, knowing that there have already been tastings in Singapore, the United States (USA) and Israel?*

Approach and scope

When conducting a risk assessment, BuRO follows the four steps according to the Codex Alimentarius, as described in microbiological risk assessement methodology: hazard identification, hazard characterisation, exposure assessment, and risk characterisation. BuRO has focused on identifying the chemical and (micro)biological agents that may be present in a cultured meat product and that may have potentially harmful health effects for humans. Animal welfare aspects, physical hazards, farmed fish and other products made from lab-grown cells have not been taken into account.

To answer the first research question, BuRO reviewed scientific literature. Details of the study carried out are described in Annex 1.

In order to answer the second question about experiences in other countries, a number of foreign food safety authorities were contacted where in vitro meat products are admitted to the market: Singapore Food Agency (SFA), the Food and Drug Administration (FDA) and the United States Department of Agriculture (USDA) Food Safety and Inspection Service (FSIS). The FDA and the USDA-FSIS are jointly responsible for supervising cultured meat production in the United States of America (USA). There have been video conferences with delegations from SFA and FDA.

The contents of this advice have been subjected to external peer review.

Findings

Legal framework

The term 'cultered meat' refers to meat products grown in vitro from animal cells. Cultured meat is a novel food in the European Union and is currently not allowed to be placed on the market. The European Commission, together with the EU Member States, decides on the authorisation of novel foods to the European market, following a food safety risk assessment by EFSA. The European regulations on novel foods have been incorporated into Dutch legislation and are enforced by the NVWA. In the Netherlands, cultured meat tastings can be organized on the basis of the CoP that has been formulated, with food safety being assessed by a dedicated expert committee specifically installed for this purpose.

Production of cultured meat

The production process of cultured meat can be divided into several steps. The first step is to obtain the starter cells for cultivation from biopsy tissue taken from an animal intended for food production. The second step is biomass production through cell growth and differentiation, where large-scale multiplication of the cells takes place, followed by differentiation (specialization) to the desired tissue type, in most cases muscle tissue.

The final step in the production process is the harvesting of cells and further processing into a meat-like end product. The cultured cell mass can be used as a raw material in food processing, with other ingredients added to provide structure, colour, aroma, and flavour. However, sometimes the product can also be harvested directly from the bioreactor and packaged without further processing. After completion of the production process, cultured meat is chilled and stored, or transported and distributed directly, after which it can be prepared for consumption.

Hazard identification

- BuRO has focused on identifying the (micro)biological and chemical hazards that can occur during the various phases of the cultured meat production process and that can have potentially harmful health effects for humans.
- Disease-causing infectious agents, including microorganisms (such as bacteria, viruses, fungi and parasites) and non-living infectious particles (such as prions), which may be present in the biopsy tissue taken from the source animal, constitute a potential microbial source of infection of cultured meat. Visual inspection of the animal and the tissue taken, and information on the health status of the animal, can provide information on potential pathogenic microorganisms present. However, it is also possible that human pathogenic microorganisms may be present in the animal without visible symptoms.
- Other potential sources of microbial contamination are culture medium additives or structural components of animal origin or produced by precision

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fermentation. Furthermore, microbial contamination may occur during tissue collection from the environment, throughout cultivation due to contaminated culture equipment, during harvesting and subsequent processing, or from the individuals carrying out these procedures. The outcome of microbial contamination in cultured meat is largely determined by the type of micro-organism and the moment of contamination. Bacterial contamination during the culture phase will in many cases terminate the culture. However, laboratory conditions may promote the growth and survival of microorganisms that are not typically associated with conventional meat production chains, but that may nevertheless cause illness if transmitted through the consumption of cultured meat.

- The potential use of antimicrobial agents during the initial phase of cultivation may contribute to the development of antimicrobial resistance. Through the consumption of cultured meat, microorganisms carrying drug resistances could be transmitted to consumers.
- Cultured meat production involves processes and microbial risks comparable to those of conventional meat production, particularly during the food processing phase. Heating and proper storage conditions are crucial steps in controlling pathogens. Existing systems such as HACCP (Hazard Analysis and Critical Control Points), GMP (Good Manufacturing Practices), GCCP (Good Cell Culture Practices), as well as hygiene and preparation guidelines, provide a useful framework for risk management, if adapted to the specific characteristics of cultured meat production.
- Chemicals may have toxic effects on a consumer of cultured meat when they remain at sufficient concentrations in the final cultured meat product. One potential source of chemical contamination is residual substances in the biopsy tissue, including environmental contaminants, residues of veterinary medicinal products and feed additives, and natural toxins. Another possible source are bioactive molecules, such as hormones, and growth and differentiation factors from the culture medium, which may have undesirable biological effects on the consumer. Furthermore, by-products of degradation or transformation processes of structural components of cultured meat may persist and have toxic effects. Materials that come into contact with the culture can also be a source of chemical contaminants, such as heavy metals from the bioreactor, microplastics from culture flasks, and residues of cleaning and disinfection agents used for cleaning tissue collection and culture equipment. In addition, antimicrobial agents that may be applied during tissue collection, and residues of cryoprotectants used during freezing of cells for storage in cell banks, may pose chemical hazards. Finally, process contaminants and substances formed through chemical transformation of cultured meat components as a result of food processing and preparation may present a potential hazard.
- Other potential hazards are the formation of toxic or allergenic proteins due to spontaneous or induced genetic instability of the cultured cells. When culture medium components or structural components are derived from a plant source or produced by precision fermentation, allergenic proteins could be present.

Hazard characterisation

- BuRO has identified potential hazards associated with the consumption of cultured meat, based on insight into the production process.
- No data are available on the actual presence of microbiological agents or chemical substances in cultured meat offered for consumption at Dutch tastings. For this reason, BuRO has not carried out any further characterisation (description of the effects of the hazards on human health) of the identified hazards.

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Exposure assessment and risk characterisation

- Cultured meat consumption at Dutch tastings involves short-term intake, with participants allowed to taste a maximum of 10 times per year according to the CoP. The portion sizes are unknown.
- For short-term intake, hazards with acute health effects are particularly relevant. In addition, non-acute effects could occur following short-term intake when, for example, a long incubation period is involved.
- For many chemical substances, effects only occur after long-term or chronic exposure. Microbiological contamination of cultured meat can lead to acute infection among participants of tastings.
- Due to the lack of data on the actual presence of pathogenic micro-organisms and levels of chemical substances in cultured meat, the (micro)biological and chemical health risks for participants in Dutch cultured meat tastings cannot be determined.

Risk assessments of novel foods and tastings in the EU and abroad

- The EU and Singapore have developed guidelines for the information that producers must submit when applying for market authorisation of a novel food. These guidelines have many similarities. All information relating to the food safety of long-term or chronic consumption of the novel food must be provided. This information is used for risk assessment. The United States does not have formal regulations on novel foods. However, the US does use an evaluation procedure when a food has not previously appeared on the US market. Risk assessments and/or evaluations for market authorisation of novel foods in the EU, Singapore and the US are carried out by food safety authorities.
- The Singapore guidelines also specifically state the information requirements that an application must meet in order to conduct tastings of novel foods. The procedure for organising tastings in the Netherlands similarly describes the information that the expert committee assessing food safety should receive regarding the composition of the cultured meat product and the safety of acute intake. The information requirements for applications to organise novel food tastings in Singapore and cultured meat tastings in the Netherlands show many similarities.
- The guidelines on information requirements for applications for market authorisation of novel foods in the EU and Singapore, as well as the guidelines for applications to organise (cultured meat) tastings in Singapore and the Netherlands, are formulated in such general terms that they do not provide sufficient insight into which specific food safety information is actually submitted by food producers. Few specific potential hazardous chemical substances or microbial contaminations are mentioned in the guidelines.

Uncertainty analysis

BuRO does not have access to the confidential content of the applications for the tastings that have already taken place in April and July 2024 and February 2025. No measurement data are available on the presence of pathogenic micro-organisms and chemical substances in cultured meat products offered by Dutch producers at tastings. Due to the limited availability of information, there is a high degree of uncertainty regarding the food safety risks of cultured meat tastings.

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Conclusion

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Cultured meat is a novel food in Europe and falls under the legislation laid down in the Novel Food Regulation (EU) 2015/2283[1]. In order to place a novel food on the market in Europe, a system has been established to safeguard food safety. This system includes a risk assessment by EFSA to determine whether consumption of the novel food poses risks to human health. A positive assessment by EFSA forms the basis for market authorisation in the EU, followed by a decision by the European Commission and the Member States on whether the product can be placed on the market. European legislation on novel foods has been incorporated into Dutch law and is enforced in the Netherlands by the NVWA. The supervision of the NVWA is based on the established food safety system as it is set up in European and Dutch laws and regulations.

In the Netherlands, a procedure is currently in place to enable cultured meat tastings, which falls outside the framework of the established food safety system. Food safety during tastings is assessed by a committee established specifically for this purpose. This committee is not affiliated with a renowned food safety institute and little is known about its composition. The information submitted by producers in an application to organise cultured meat tastings, as well as the accompanying safety dossier, is confidential and not publicly accessible, and therefore also unknown to the supervisory authority. The requirements for submitting an application to organise cultured meat tastings also do not provide sufficient insight into which food safety information is actually provided by cultured meat producers. It is undesirable and ineffective that the supervisory authority has no insight into the safety assessment of products that fall under its monitoring responsibility. The food safety assessment procedure during cultured meat tastings is not transparent to the supervisory authority and the reproducibility of the process is uncertain.

Based on insight into the production process, BuRO has identified potential hazards associated with the consumption of cultured meat. However, no data are available on the actual presence of microbiological agents or chemical substances in cultured meat offered for consumption at Dutch tastings. Because these essential data are lacking, the health risks for participants in cultured meat tastings cannot be assessed. It therefore cannot be ruled out that risks are associated with the consumption of cultured meat at tastings held in the Netherlands. Due to the lack of transparency in the assessment process, it is unknown whether the potential food safety risks are adequately addressed and whether they are assessed in a sound and reproducible manner under the current procedure established for this purpose.

Answer to the questions

Question 1: *What chemical and (micro)biological risks are associated with tasting cultured meat?*

Microbiological hazards that may pose a potential risk include contamination with pathogenic microorganisms originating from the source animal, from structural or medium components, from contaminated equipment, or from human handling during the cultivation process, and/or during subsequent processing and preparation. In addition, spontaneous or induced genetic instability of the cultured cells may result in the formation of toxic or allergenic proteins.

Chemical hazards that may pose a potential risk include contaminants in the biopsy tissue, potentially toxic residues or metabolites of structural or medium components, bioactive molecules in the culture medium, and chemical

[1] Regulation (EU) 2015/2283 of the European Parliament and of the Council of 25 November 2015 on novel foods

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contaminants from contact materials, including cleaning and disinfection agents, cryoprotectants, and heavy metals and microplastics.

BuRO does not have access to information on the exact production process, the origin of the cells, the ingredients of the culture medium, or the nature of the structural components of cultured meat intended for Dutch tastings. Nothing is known either about the storage and preservation conditions. In the case of tastings, cultured meat consumption involves short-term intake. The (micro)biological and chemical hazards that pose a potential risk therefore mainly concern agents that may have an acute effect on the health of tasting participants. In addition, non-acute effects may occur after short-term intake if, for example, a long incubation period is involved. Data on the actual presence of pathogenic microorganisms and chemical substances in cultured meat are lacking. As a result, the (micro)biological and chemical health risks for participants in Dutch cultured meat tastings cannot be determined.

Question 2: *What (scientific) knowledge has already been gained abroad, knowing that tastings have already taken place in Singapore, the United States (US), and Israel?*

In Singapore, tastings of novel foods are only permitted as part of product development. To organize tastings, an exemption must be requested from the SFA. For this, a risk assessment must be conducted for acute hazards related to microbiological agents, contaminants, toxins, and allergens that may potentially be present in the product to be tasted. It is the producer's responsibility to demonstrate that short-term intake of the novel food does not cause adverse health effects and that the tastings can be carried out safely. The SFA has received no reports of adverse health effects following tastings of cultured meat.

In the US, tastings of novel foods for commercial purposes are not permitted. Tastings for research purposes are not formally prohibited. Only employees of the producer or contracted test subjects may participate in such tastings. It is the producer's responsibility to demonstrate that the product to be tasted is safe for human consumption. The FDA and USDA-FSIS jointly oversee food made from animal cells. Every food producer must undergo an inspection by these authorities to assess compliance with legal food safety requirements. The FDA and USDA-FSIS have received no reports of adverse health effects following tastings of cultured meat.

Contact with foreign food safety authorities SFA and FDA did not provide additional information regarding the potential chemical and (micro)biological risks associated with tastings of cultured meat.

Advice from BuRO

To the minister of Health, Welfare and Sport

- Provide the NVWA with access to the safety assessments of cultured meat tastings carried out by the committee established specifically for this purpose.
- Ensure the safeguarding of food safety during tastings of novel foods. To this end, establish an assessment system that includes the requirements for submitting an application to organize tastings, the accompanying safety dossier, and the authority that evaluates it, which meets the following criteria:
  - embedded in the established and proven food safety systems as set out in European and Dutch legislation and regulations;
  - accessible to the supervisory authority;
  - transparent and reproducible.

To the Inspector-General of the NVWA

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- Advocate the safeguarding of food safety during tastings of novel foods, including cultured meat, which is accessible to the NVWA.

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17 July 2025

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Yours sincerely,

Prof. dr. Dick T.H.M. Sijm
Director of the Risk Assessment & Research

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Substantiation

Advice on the public health risks associated with offering cultured meat at tastings

1 Introduction

The production of cultured meat is a form of cellular agriculture, a collective term for technologies used to produce products such as milk and meat directly from animal and/or microbial cells (Rischer et al., 2020). The term 'cultured meat' refers to meat products grown from animal cells under laboratory conditions (also referred to as in vitro)(Bryant, 2020). In contrast, conventional or traditional meat is harvested from slaughtered animals. It is estimated that cells from a single source animal can be used to produce much more meat than the amount obtained from a single carcass (Connor, 2013). In addition, estimates suggest that large-scale industrial production of cultured meat may require less land and water and cause lower greenhouse gas emissions than the conventional meat industry (Tuomisto & de Mattos, 2011; Mattick et al., 2015).

In 2013, the first in vitro cultured beef hamburger was presented by Dutch researchers[2]. In December 2020, the first cultured chicken nuggets were approved for market entry in Singapore, consisting of a mixture of cultured chicken meat and plant-based ingredients (Waltz, 2021). This made Singapore the first country to formally authorize the consumption of in vitro cultured meat[3]. According to the Singapore Food Agency (SFA), the product met all quality and safety requirements for conventional chicken meat. More recently, the sale of in vitro cultured chicken has also been approved in the United States by the United States Department of Agriculture (USDA) and the Food and Drug Administration (FDA)[4]. In January 2024, the sale of cultured beef was authorized by the Israeli government[5]. In 2021, more than 100 start-up companies worldwide were engaged in developing various cultured meat products (Cohen et al., 2022).

1.1 Legal framework

In Europe, cultured meat is regarded as a novel food and must comply with the European Novel Food Regulation (EU) 2015/2283[6]. Within the EU, novel foods are defined as "any food that was not used for human consumption to a significant degree within the Union before 15 May 1997" and that falls under at least one of the specified categories. These categories also include "foods consisting of, isolated from, or produced from cell culture or tissue culture derived from animals, plants, microorganisms, fungi, or algae." When genetic modification has been applied, Regulation (EC) No. 1829/2003 also applies[7]. A manufacturer wishing to place a cultured meat product on the European market must submit an application for authorization to the European Commission (EC). The applicant must provide a scientific dossier with product information to the EC. At the request of the EC, the European Food Safety Authority (EFSA) conducts a risk assessment based on the submitted dossier. The final decision on authorizing a cultured meat product for the European market is made by the EC and the EU Member States, following EFSA's advice on food safety. Once approved, the novel food is added to the Union list of authorized novel foods.

European regulations apply directly in all Member States, including the Netherlands. In the Netherlands, the European legislation on novel foods has been implemented through the Decree on novel foods and genetically modified foods (Warenwetbesluit nieuwe voedingsmiddelen en genetisch gemodificeerde levensmiddelen)[8]. This decree covers two main aspects:

1. Prohibition: it is forbidden to place a novel food on the market in the Netherlands if it is not on the Union list or if it does not meet the conditions and specifications stated in that list.

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[2] World's first lab-grown burger is eaten in London - BBC News

[3] Singapore Gives Regulatory Approval for Eat Just's Cell-Based Chicken

[4] U.S. approves sale of cell-cultured chicken to two startups

[5] Israel approves sale of Aleph Farms' cultivated beef | Food Safety News

[6] Regulation (EU) 2015/2283 of the European Parliament and of the Council of 25 November 2015 on novel foods

[7] Regulation (EC) No 1829/2003 of the European Parliament and of the Council of 22 September 2003 on genetically modified food and feed

[8] wetten.nl - Regeling - Warenwetbesluit nieuwe voedingsmiddelen en genetisch gemodificeerde levensmiddelen - BWBR0040236

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2. Designation of the competent authority: producers may consult the Minister of Health, Welfare and Sport (VWS) to determine whether a food falls under Regulation (EU) 2015/2283, after having first reviewed all available information themselves. The Minister takes a decision on this matter. The details of the consultation procedure are set out in the Regulation on the consultation procedure for novel foods (Warenwetregeling raadplegingsprocedure nieuwe voedingsmiddelen)[9].

A producer may submit a consultation request with a technical dossier to the Novel Foods Unit of the Medicines Evaluation Board. Such a request is used to determine whether a product falls under novel food legislation. The Novel Foods Unit of the Medicines Evaluation Board advises the Minister of VWS accordingly.

The Netherlands Food and Consumer Product Safety Authority (NVWA) is responsible for enforcing novel food legislation. A food business operator is responsible for demonstrating compliance with this legislation. During an inspection, the NVWA may request and assess the producer's evidence. Any novel food not included on the Union list is considered potentially harmful by the NVWA, as no risk assessment has been conducted. If novel foods are marketed that are not included on the Union list, or if an authorized novel food does not comply with the applicable conditions and specifications, the NVWA may take enforcement action, issue warnings or fines, or require the producer to withdraw the product from the market.

1.2 Cultured meat tastings in the Netherlands

The organization of cultured meat tastings in the Netherlands falls outside the legal framework described above. In March 2022, a motion submitted by Members of Parliament Tjeerd de Groot and Valstar (Parliamentary Paper 27 428, No. 383) was adopted by the House of Representatives, requesting the government to consult with Dutch producers of cultured meat to allow tastings to take place under "controlled and safe conditions." Following this, cultured meat producers and the Dutch association of biotechnology companies (HollandBio), in consultation with the Ministry of Health, Welfare and Sport (VWS) and the Ministry of Agriculture, Fisheries, Food Security and Nature (LVVN), developed a Code of Practice (CoP) for organizing cultured meat tastings.

The CoP specifies the information that must be included in an application submitted by a cultured meat producer to organize tasting sessions. It also describes the procedure for evaluating such an application. An expert committee, consisting of at least a toxicologist, a microbiologist, a physician, and an ethicist, assesses whether a tasting may safely take place based on the submitted application. The committee's decision must be unanimous. The committee was established by the Cellular Agriculture Netherlands Foundation (CANS)[10], founded in 2021 by entrepreneurs, scientists, pioneers, and food enthusiasts. According to its stated objectives, the foundation aims to "promote cellular agriculture as a means to make the food system more sustainable" and "at the same time provide people with delicious, nutritious, and safe meat and dairy products." In 2022, CANS received financial support from the Dutch government.

The CoP also describes the procedural conditions that tastings must meet. Participation in tastings is strictly voluntary and limited to healthy adults without underlying conditions or allergies, and only after obtaining informed consent. Participants are not reimbursed. The number of participants is limited to 30 per tasting, and tastings are not publicly accessible. Any adverse health effects are monitored for up to two weeks after the tasting.

In April and July 2024, two Dutch producers of cultured meat organized the first tastings in the Netherlands, inviting pre-selected groups to sample their cultured meat products.

2 Cultured meat production process

The term "meat" generally refers to an exsanguinated and dehydrated product derived from the skeletal muscle system of animals slaughtered for this purpose (Lautenschlaeger & Upmann, 2017). Similar to conventional meat, a cultured meat product primarily consists of edible muscle tissue. The cultivation of muscle tissue focuses mainly on the proliferation of myoblasts, the precursor cells of muscle fibers, which fuse into mature muscle cells (myocytes or myofibers) (Wood et al., 2023). The addition of adipose tissue has been shown to be important for the flavor, texture, and overall eating experience of (cultured) meat (Fish et al., 2020; Dohmen et al., 2022; Pallaoro et al., 2023).

[9] wetten.nl - Regeling - Warenwetregeling raadplegingsprocedure nieuwe voedingsmiddelen - BWBR0041262

[10] Cellulaire Agricultuur Nederland

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The production process of cultured meat may vary depending on the type of cells used as starting material and the type of product being developed. In general, the process consists of several steps: 1. Obtaining the starting cells, 2. Biomass production through cell growth and differentiation, 3. Harvesting the cells and subsequent food processing. These steps are described below.

1. *Obtaining the starting cells*
   The production process of cultured meat begins with the selection of the source animal (currently most often cattle, pigs, or chickens) and the source tissue from which the starting cells are obtained (Chodkowska et al., 2022). Cells may be collected from freshly slaughtered animals or from living animals through a biopsy (Melzener et al., 2021). The harvested tissue typically consists of a mixture of different cell types, from which the desired cell type can be further enriched. Various cell types can be used as starting material for the cultivation process. Embryonic stem cells, induced pluripotent stem cells, mesenchymal stem cells, and muscle stem cells (satellite cells; precursors of myoblasts) are commonly selected as initial material for the cultivation of muscle or adipose tissue (Stephens et al., 2018; Choi et al., 2021; Reiss et al., 2021). Stem cells possess the unique ability to self-renew and to differentiate into other cell types (Nava et al., 2012). Embryonic stem cells are also referred to as pluripotent stem cells, as they are capable of differentiating into any type of body cell (Liang & Zhang, 2013). Induced pluripotent stem cells are somatic cells that can be reprogrammed into pluripotent cells using specific technologies (Takahashi & Yamanaka, 2006). Mesenchymal stem cells are multipotent cells capable of differentiating into a number of cell types, including bone, cartilage, muscle, and adipose cells (Ding et al., 2011). Satellite cells are the resident stem cells of skeletal muscle (Wang et al., 2014).

2. *Biomass production through cell growth and differentiation*
   In the second step of the cultivation process, the isolated cells are expanded on a large scale, followed by the differentiation, or specialization, of these cells into muscle and/or adipose tissue. To form a cultured meat product, a large quantity of differentiated muscle cells is ultimately required. However, most cells have a limit to the number of divisions they can undergo (Shay & Wright, 2000). Once this maximum is reached, cell proliferation ceases. In general, the further a cell type is differentiated, the lower its proliferative capacity (Ruijtenberg & van den Heuvel, 2016). Consequently, when cell types that are more differentiated toward muscle cells are used, it may be challenging to obtain sufficient numbers of cells (Zhang et al., 2020). Embryonic stem cells can divide indefinitely (Amit et al., 2000). However, in vitro differentiation of these cells into muscle cells is a complex process (Jiwlawat et al., 2018). Prior to scaling up, cells with favourable properties for the cultivation process, such as extended lifespan, increased proliferative potential, or resilience under specific conditions, may be selected (Reiss et al., 2021). To obtain cells with the desired traits, multiple rounds of selection may be applied, during which cells with advantageous spontaneous genetic variations are retained (Reiss et al., 2021). Alternatively, cells can be actively modified to acquire beneficial properties using genetic or chemical modification techniques (Ramboer et al., 2014).

   Cultivation takes place under sterile conditions, beginning in a plastic culture flask and progressing to a bioreactor of 1,000–10,000 liters at later stages (Kulus et al., 2023). In bioreactors, conditions critical for cell growth, such as temperature, oxygen levels, and supply of culture medium, are regulated. Additional parameters such as pH, metabolite formation, and the extent of biomass accumulation can also be monitored. The requirements of the cultivation environment vary depending on the growth stage and the cell type used. The composition of the culture medium is also of major importance. During the proliferation stage, in which cell numbers increase exponentially, a growth medium primarily contains nutrients and factors required for cell expansion, such as glucose, amino acids, fatty acids, inorganic salts, vitamins and minerals, as well as hormones, signalling molecules, and growth factors (Chriki & Hocquette, 2020; O'Neill et al., 2022). After the proliferation stage, stem cells must differentiate into the desired cell type, such as muscle cells. Differentiation can be induced by altering the composition of the culture medium, often in combination with adjustments to other cultivation conditions and mechanical stimulation (Li et al., 2021).

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Cells used for cultured meat production are typically adherent cells, meaning that in the *in vivo* situation they require attachment to a surface. This surface is provided by the extracellular matrix (ECM), an environment outside the cell composed of fibrous proteins and long sugar chains that provide structure and strength to tissues. In bioreactors, however, cultured cells need to grow in suspension. To achieve this, bioreactors often employ so-called microcarriers. These are small beads (approximately 100–400 μm in diameter) to which cells can attach, as the beads mimic the stiffness, porosity, and surface characteristics of the ECM. It is also possible to grow cells on a more complex scaffold. A scaffold is a three-dimensional structure that supports growth, maturation, and differentiation of the culture, facilitating the formation of more complex structures such as multilayered arrangements of cells. Through scaffolds, oxygen and nutrients can be delivered to the growing cell mass, comparable to the vascularization of tissues in living animals. Cells can also be supplied with growth and differentiation factors, or with molecules influencing the flavour or colour of the final product. Using a scaffold, cells can further be stimulated mechanically or electrically to grow into (muscle) fibre-like structures. Mechanical stimulation can, for example, be provided by embedding cells in scaffold material with specific stiffness, which promotes differentiation. It has been demonstrated that the differentiation of mesenchymal stem cells into fat or bone can be directed by the stiffness of the scaffold material. Scaffolds or microcarriers can be made from natural or synthetic materials, or combinations thereof. These materials may or may not be edible or degradable, and may or may not need to be removed later in the production process.

3. *Harvesting of the Cells and Further Food Processing*
   When the cell culture has reached its maximum growth density and has differentiated into the desired cell type, the cells must be harvested and collected. This may also involve detaching the cells from microcarriers or a scaffold, where it is important that the cells are not damaged. Various dissociation methods can be used for this purpose, such as enzymatic or chemical treatments (Bodiou et al., 2020). Once harvested, the cells will die due to the cessation of nutrient supply. In conventional meat production, cell death occurs when an animal is slaughtered. This death is part of the process in which muscle tissue is transformed into meat (Fraeye et al., 2020). This is a complex biochemical process involving major metabolic changes, which has not yet been fully elucidated (Fraeye et al., 2020). It remains unclear whether and how a comparable transformation occurs during the formation of cultured meat from cultivated cells. Following harvesting, further processing steps are undertaken to produce a product resembling meat. For some products that have been grown on a scaffold, the product can be harvested directly from the bioreactor and packaged without additional processing. For other products, the harvested cell mass can serve as raw material for processing steps similar to those used in conventional meat or plant-based meat alternatives. In yet other cases, the final product is shaped using 3D-printing techniques or other methods. This may involve combining different cell types (such as muscle and fat tissue) and adding other components and ingredients to improve colour, flavour, and aroma, as well as applying techniques to provide structure and texture to the product (Fraeye et al., 2020; Wood et al., 2023). The precise methods for cell harvesting and processing into the final product are largely not publicly disclosed. The total culture duration until formation of the end product generally takes several weeks, depending on the type of product being produced.

For large-scale and continuous production of cultured meat, there are broadly two options. In one option, cells must be collected from animals on a regular basis (Stephens et al., 2018; Melzener et al., 2021). Cells that are directly isolated from animal tissue and placed into culture are referred to as primary cells. Primary cells can only undergo a limited number of divisions in vitro before they age and enter a quiescent state, which necessitates repeated collection of new cells from animals (Hayflick & Moorhead, 1961; Di Micco et al., 2021). In the other option, cells are either selected after spontaneous immortalization, or are immortalized through chemical or genetic induction, after which they can theoretically continue dividing indefinitely (Stephens et al., 2018; Soice & Johnston, 2021). This approach leads to the establishment of stable cell lines that retain their genetic and physiological characteristics, thereby enabling consistent and uniform meat production (Specht et al., 2018). To this end, cells can be preserved at various stages of the culture process by cryopreservation. In this way, a cell bank can be created, from which individual aliquots of identical cells can be thawed to initiate cultures, ultimately yielding comparable cultured meat products (Sun et al., 2016).

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3 Hazard identification

For this hazard identification, the possible chemical and (micro)biological hazards for the consumer that may arise with the production of cultured meat have been mapped out. The hazards identified by EFSA and FAO & WHO have been included (FAO & WHO, 2023; EFSA et al., 2024). Specific attention was given to materials, ingredients, and processes that are unique to cultured meat production, such as the use of scaffolds, (genetic) modification of cellular properties, and supplementation via the culture medium. Table 1 provides an overview of the identified hazards. The hazards are classified into chemical, microbiological, and other hazards.

Microbiological contamination during the cultured meat production process can occur at different stages and via various routes. Pathogens are microbiological agents such as bacteria, viruses, fungi, parasites, and prions (infectious proteins) that can cause disease in humans through infection. Chemical hazards are (residues of) chemical substances that may have toxic effects on a consumer of cultured meat. Bioactive molecules such as hormones, and growth and differentiation factors, are included in this assessment.

Below, the identified chemical, (micro)biological, and other hazards associated with cultured meat are described in more detail for each production step.

3.1 Hazards associated with obtaining the starting cells

This section describes the microbiological, chemical, and genetic hazards (listed under "Other hazards" in Table 1) related to the first step in the production process of cultured meat: obtaining the starting cells. These include hazards associated with the source animal, hazards that arise or occur during the collection of the tissue that serves as starting material for a culture, and hazards resulting from further cell selection, as well as the formation of cell lines and cell banks.

3.1.1 Microbiological hazards

Microbiological hazards associated with obtaining the starting cells include pathogens present in the source animal and microbial contamination during tissue collection.

The cells used for cultured meat production are taken from either recently slaughtered animals or from live animals through a biopsy. If the source animal is infected with pathogenic microorganisms, and these microorganisms are present in the collected tissue, this can pose a potential hazard to the health of cultured meat consumers. Pathogenic microorganisms that may be present in or on the source animal include bacteria, viruses, fungi, and parasites. The source tissue may also be contaminated with prions or endotoxins (bacterial cell wall residues).

Many foodborne infections are caused by bacteria such as *Salmonella* spp., *Escherichia coli*, and *Campylobacter*, which inhabit the intestinal tract of animals. In conventional meat production, contamination mainly occurs during slaughter, when contact with intestinal contents, faeces, or contaminated hides leads to bacterial contamination of the carcass. Since whole carcasses are not processed in the production of cultured meat, the risk of contamination through this route is greatly reduced. However, contamination with these bacteria can still occur during the collection of tissue used to initiate a cell culture (Habowski & Sant'Ana, 2024; Powell et al., 2025).

In addition to these common zoonotic enteric bacteria, contamination could also occur with pathogenic microorganisms for which specific animal species serve as reservoirs, such as *Brucella abortus* and *Mycobacterium bovis* (Habowski & Sant'Ana, 2024; Powell et al., 2025). However, the prevalence of these microorganisms in Dutch cattle is extremely low and occurs only occasionally, mainly in imported animals.

The source animal may also carry viral pathogens. For a virus to present a hazard to the eventual cultured meat consumer, it must be able to cross the species barrier between animal and human, and remain viable and capable of replication during the culture phase. Cattle may potentially be infected with the species-specific Bovine Leukemia Virus (BLV). Although not yet conclusively demonstrated, the scientific literature suggests that BLV could be zoonotic and foodborne (Buehring et al., 2019). However, it is still unknown whether BLV can indeed act as a human pathogen. Likewise, it is unclear whether the virus could survive and replicate during the in vitro culture phase of cultured meat production.

Foodborne parasites such as *Toxoplasma gondii* and *Cryptosporidium parvum* represent potential hazards that may enter cultured meat through a contaminated source animal (Habowski & Sant'Ana, 2024; Powell et al., 2025).

An infectious prion is a misfolded protein capable of transmitting its abnormal structure to other copies of the same protein. Prions are particularly present in brain, spinal cord, lymph nodes, and

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Table 1 Hazard identification of cultured meat.

Process step	Sub-process step	Chemical	Microbiological	Other
1. Obtaining the starting cells	i. Biopsy	Presence in biopsy tissue of: Environmental contaminants, Residues of veterinary medicinal products Residues of feed additives Natural toxins Residues of cleaning and disinfection agents Antimicrobial agents	Pathogenic micro-organisms from the source animal Microbial contamination during tissue collection Antimicrobial resistance	
	ii. Cell isolation and establishment of stable cell lines (including storage)	Residues of cryoprotectants		Genetic modification (allergens, toxins) Genetic instability (allergens, toxins)
2. Biomass production (cell growth and differentiation)	i. Culture medium (components)	Bioactive molecules (hormones, growth- and differentiation factors) Residues of metabolites (toxic)	Animal source: Contamination with pathogenic micro-organisms Precision fermentation: Microbial contamination	Plant source: Allergens Precision fermentation: Allergens
	ii. Scaffold	Residues of metabolites (toxic) Toxic byproducts of transformation processes	Animal source: Contamination with pathogenic micro-organisms Precision fermentation: Microbial contamination	Plant source: Allergens Precision fermentation: Allergens
	iii. Bioreactor (and other instruments)	Heavy metals Microplastics Residues of cleaning- and disinfection agents	Microbial contamination (including toxins)	
3. Cell harvest and further processing		Process contaminants Toxic substances through chemical transformation	Microbial contamination	

parts of the intestines (Gough & Maddison, 2010). Avoiding these tissues when sourcing starting cells for cultured meat production, and monitoring source animals for clinical symptoms of prion-induced neurodegenerative diseases, is essential to prevent prion transmission through cultured meat.

To control the risk of transmitting pathogens from the source animal to the cultivated meat consumer, it is important that the harvested tissue is thoroughly inspected (Ong et al., 2021; FAO & WHO, 2023; Gu et al., 2023). Documented information about the health status of the source animal may also provide indications of potentially present pathogenic microorganisms. Visual inspection of the animal and the harvested tissues can reduce the likelihood of transmission of diseases such as brucellosis and tuberculosis (EFSA, 2013). However, not all human pathogens cause clinical signs in animals. As a result, an asymptomatic animal and tissue that appears healthy may still carry microorganisms that can cause disease in humans.

Since the collection of cells or tissues from living or slaughtered animals cannot take place under sterile conditions, there is a risk of contamination of the biopsy tissue during this procedure. The source of contamination can be microorganisms from the environment, from the person performing the procedure, or from contaminated instruments. Because sterility can be poorly guaranteed, many cultured meat producers apply antimicrobial agents during tissue collection and/or at the start of the cell culture. The use of antimicrobial agents can induce resistance in

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microorganisms. This may occur due to underdosing and/or repeated exposure. A possible consequence is that microorganisms with resistances may end up in cultured meat consumed by humans. If these microorganisms cause disease in the consumer, or if the resistance is transferred to pathogenic microorganisms already present in the consumer's body, this may lead to reduced effectiveness of treatments against bacterial infections.

Pathogens originating from the source animal itself, or pathogens introduced during the tissue collection process, can pose a risk to the cultured meat consumer. For this to happen, microorganisms must be able to survive the culture phase and remain in the final product in concentrations high enough to be harmful to the consumer. In addition, they must withstand food processing and preparation steps, such as heating. Depending on the type of microorganism, microbial contamination can render the initiation of a cell culture unfeasible. This is described in more detail in section 3.2.1 on microbial contamination during the culture phase.

3.1.2 Chemical hazards

Chemical hazards associated with the procurement of starter cells include the presence of environmental contaminants, residues of veterinary drugs and feed additives, and natural toxins in the biopsy tissue. Other hazards include residues of cleaning and disinfecting agents used to sterilize the instruments for tissue collection, antimicrobial agents applied during tissue harvesting, and residues of cryoprotectants (FAO & WHO, 2023).

An animal intended for meat production may be exposed to contaminants and residues originating, for example, from the environment or from animal feed. Environmental contaminants are chemical substances that mainly enter the environment as a result of industrial activities and/or combustion processes. The cells or tissues taken from livestock to serve as starting material for cell culture may be contaminated in the same way as conventional meat with chemical substances from the environment and/or feed, or through treatments with veterinary drugs such as NSAIDs, antiparasitic agents, and antibiotics. Natural toxins, such as plant and mycotoxins, may also be transferred into food via meat. Unlike conventional meat, however, the cultured meat production process takes place before the animal tissue ultimately reaches the consumer as a final product. At present, it is unclear what effect the cultivation process has on the chemical substances that may occur in animal tissue. It is plausible that chemical substances present in biopsy tissue are diluted during cultivation.

Residues of cleaning and/or disinfecting agents may remain on the instruments used to collect cell tissue from the source animal. Through this route, the starting tissue of the culture can become contaminated with these substances. Antimicrobial agents such as antibiotics and fungicides may also be applied during tissue collection, leaving residues behind in the culture. During the cell proliferation and differentiation phases, most producers no longer use antimicrobial agents. Therefore, it is likely that these substances are diluted to very low concentrations by the end of the culture. Nevertheless, residues of these substances may still remain in the final product. Consumer exposure to residual antimicrobial agents can have undesirable effects, such as disruption of gut microbiota and the possible development of antimicrobial resistance (Okocha et al., 2018). Antimicrobial resistance may result in diseases becoming more difficult to treat (Salam et al., 2023).

When immortalized cell lines are frozen for storage in cell banks, substances are used to protect cells against the adverse effects of freezing, such as crystal formation (Woods et al., 2016; Elliott et al., 2017). These substances are referred to as cryoprotectants. Commonly used cryoprotectants include dimethyl sulfoxide, polyethylene glycol, and sucrose. When cells are thawed to initiate a new culture, residues of these cryoprotectants may remain in the culture. Since these substances are not applied during subsequent cultivation, they become diluted to low concentrations. However, due to the potential toxicity of these substances, it is important to determine the concentration of residues in the final product (Woods et al., 2016; Ong et al., 2021; FAO & WHO, 2023).

Chemical substances present in the starting tissue of the culture could potentially cause adverse health effects to a consumer of cultivated meat. For this to occur, these substances or any harmful metabolites must persist throughout cultivation and remain in the final product at sufficiently high concentrations to be harmful to the consumer.

3.1.3 Genetic instability and carcinogenicity

In addition to chemical and microbiological hazards associated with the procurement of starter cells for cultured meat production, a potential hazard for the cultured meat consumer may also arise from spontaneous or induced genetic alterations in the cells.

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To produce a cultivated meat product, a large number of cells is required. To reach these quantities, cells must undergo many rounds of proliferation. It is therefore in the interest of cultured meat producers to ensure that the cells retain their proliferative capacity and can continue dividing beyond their intrinsic limit. This can be achieved through "spontaneous immortalization," in which cells, due to unregulated adaptation to in vitro culture conditions, accumulate mutations (genetic changes) that allow them to continue multiplying. However, immortalization can also be achieved in a more directed manner. Cultured meat producers may apply genetic modification to introduce mutations that cause cells to lose unfavourable traits for the cultivation process or gain favourable ones, such as unlimited proliferation (Soice & Johnston, 2021). Genetic modification encompasses a range of techniques by which the DNA of an organism can be altered.

Some genetic modification techniques involve integration into the genome of the cells. This ensures that the alterations are passed on from the mother cell to its progeny during cell division. These alterations will therefore also be part of the cells that form the final cultured meat product. However, non-integrative genetic modifications are also possible, which will not be present in the final product. Furthermore, alternatives to genetic modification can be used to support the proliferative capacity of cells, such as treatment with proteins or small molecules. Another strategy is to mimic conditions that stimulate cellular proliferation, such as a state of tissue damage or an embryonic state. At present, it is unclear which forms of genetic modification are applied by cultured meat producers in the Netherlands.

A common feature of the various cultivation strategies employed is that cells are cultured for extended periods under laboratory conditions. As a result, they are exposed to selective pressures that can cause genetic drift (Jaime-Rodríguez et al., 2023). Genetic drift arises as mutations accumulate, ultimately leading to changes in the phenotype (observable properties or characteristics) (Ben-David et al., 2018). A conceivable effect of genetic drift is the activation of oncogenes, which are genes that can promote the development of cancer (Soice & Johnston, 2021; Jaime-Rodríguez et al., 2023). In addition to genetic drift, genetic modification may also give rise to genetic instability. For example, the insertion of an extra piece of DNA into the genome, or the deletion of a segment, may cause unintended alterations in the genetic information located within that chromosomal region. The genetic instability of cultured cells used for cultured meat production has not yet been described in the scientific literature. However, instability of in vitro cell cultures for other purposes is well documented (Ben-David et al., 2018).

Whether the genetic instability of in vitro cell cultures poses a risk to human health when these cells are consumed has not yet been sufficiently investigated scientifically (Ong et al., 2021; Smith-Uchotski & Wanjiru, 2022; FAO & WHO, 2023; Gu et al., 2023). A possible consequence of genetic instability is that protein alterations occur, that proteins are expressed at different levels, or even that new proteins or metabolites are generated (Soice & Johnston, 2021). When these changes result in molecules with toxic or allergenic properties, this represents a potential risk to food safety (Ong et al., 2021; Smith-Uchotski & Wanjiru, 2022; FAO & WHO, 2023; Gu et al., 2023).

The presence of unstable or modified DNA in cultured meat may raise consumer concerns about the possible effects of this genetic material. Conventional food products, such as meat and crops, also contain genetic material (Jonas et al., 2001). However, this is largely degraded during food processing and digestion (Jonas et al., 2001). Processes such as heating, changes in acidity, mechanical forces that break down food, and enzymatic activity ensure that DNA is broken down into small fragments, and the building blocks of DNA can then be reused by the human body (Jonas et al., 2001). Nevertheless, there is evidence that DNA fragments from food, consisting of several hundred base pairs, can be detected in the intestines (Hohlweg & Doerfler, 2001; Forsman et al., 2003). These fragments may be taken up by intestinal cells or by microbiota in the gut and can enter the circulation (Forsman et al., 2003; Spisák et al., 2013). However, there is no documented evidence that the presence of DNA from food products has clinical consequences. For a clinical effect to occur, a sufficiently long and intact DNA sequence would have to integrate into the genome. This integration would also need to occur in such a way that the fragment encodes a functional protein that is expressed and exerts a harmful health effect. In addition, integration would need to occur at a sufficient scale to produce substantial amounts of the protein. Scientifically, this appears to be a highly unlikely chain of events. Only limited studies, with conflicting results, have examined the integration of DNA fragments from genetically modified crops in the human body (Netherwood et al., 2004; Johannessen et al., 2013; Nawaz et al., 2019). Nothing is known about the uptake of genetic material from genetically modified or spontaneously immortalized in vitro cultured animal cells.

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The cells that make up a cultured meat product cannot proliferate in the body of a consumer, as the cells are dead at the time of meat consumption. Therefore, there is no risk of tumour development from eating cells that have undergone uncontrolled proliferation (Lanzoni et al., 2022; FAO & WHO, 2023). To determine whether it is safe to consume cultured meat products, it must be definitively ruled out that any harmful DNA fragments present can remain intact and integrate into the genome of consumer cells, thereby inducing genetic changes (Ong et al., 2021; Soice & Johnston, 2021). In addition, it must be confirmed that genetic instability does not lead to the formation of molecules in the cultured meat product that are allergenic or toxic.

3.2 Hazards associated with biomass production through cell growth and differentiation

This section describes the microbiological and chemical hazards associated with the second step of cultured meat production, in which isolated cells are expanded on a large scale, followed by differentiation into muscle and/or fat tissue. Hazards are identified that are related to the culture medium, scaffolds, and the bioreactor used during this cultivation phase.

3.2.1 Microbiological hazards

Microbiological hazards that can occur during the proliferation and differentiation phase of the culture include microbial contamination of medium components and scaffolds, as well as contamination of the bioreactor and/or other culture equipment.

The cultivation of animal cells is extremely vulnerable to bacterial contamination, as bacteria grow at much higher rates than animal cells (Olenic & Thorrez, 2023). This vulnerability makes it essential that in vitro culture is carried out under completely sterile conditions. Ensuring sterility throughout the cultivation phase, however, is not straightforward. Depending on the type of microorganism and the timing of contamination, such an event can either disrupt and terminate the culture, or result in a finished cultured meat product that is contaminated. In many cases, bacterial contamination will visibly disrupt the culture before a final product can be formed.

In research laboratories, contamination of cell cultures with mycoplasma species is common. Literature estimates suggest that 5–35% of cell lines are contaminated with one or more mycoplasma species (Nikfarjam & Farzaneh, 2012). Mycoplasma are small Gram-negative bacteria lacking a cell wall and are resistant to many antibiotics. Mycoplasma contamination does not cause turbidity in culture medium, making visual detection difficult. Mycoplasma can affect the growth rate of cultures, alter cell metabolism, or induce chromosomal abnormalities (Drexler & Uphoff, 2002). It remains unclear whether mycoplasma pose a direct hazard to consumers as foodborne pathogens. Of the more than one hundred known species, only a few can infect humans. Many mycoplasma species are commensals of the respiratory or urogenital tract but may act as opportunistic pathogens. *M. pneumoniae* is the best-known human pathogen, causing upper respiratory tract infections (Kashyap & Sarkar, 2010). However, no foodborne outbreaks caused by *M. pneumoniae* have been reported, and faecal–oral transmission has not been demonstrated.

Laboratory conditions, such as the culture medium composition or oxygen and carbon dioxide concentrations, could selectively support the survival and/or growth of specific mycoplasma species or other microorganisms. If these microorganisms are transmitted through cultured meat, they could potentially cause illness in consumers. It is therefore possible that pathogens relevant for cultured meat production differ from those of concern in conventional meat production chains.

The source of culture medium components, or the way they are produced, can introduce microbiological hazards to cultured meat consumers. Animal serum (such as foetal bovine serum, FBS) has historically been widely used as a nutrient source in culture media for biomedical research, as it supports the growth of a wide range of mammalian cells (Zhang et al., 2020; Skrivergaard et al., 2023). Serum is a protein-rich and highly complex mixture, containing growth factors, hormones, amino acids, vitamins, metabolites, and nutrients (Post et al., 2020). However, serum can also act as a source of microbial contaminants, including bacteria, viruses, infectious prions, and endotoxins (Hadi & Brightwell, 2021; Ong et al., 2021). Scientific studies have shown that 20–50% of commercially available FBS samples contain viruses (Erickson et al., 1991; Wessman & Levings, 1999). For example, the virus responsible for bovine viral diarrhoea is frequently detected in FBS and can be propagated in cell cultures during vaccine production (Laassri et al., 2018). This highlights the need for systematic detection of microbial contaminants in culture medium components. Zoonotic viruses such as influenza, rotavirus, hepatitis A, and hepatitis E may be present in animal-derived components and could theoretically be transmitted to humans through cultured meat.

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In addition to the potential presence of microorganisms, serum composition can vary considerably, and the use of serum is inconsistent with the goal of producing meat without the need to slaughter animals. For these reasons, many cultured meat producers rely on serum-free media and seek alternative sources of medium supplements.

When culture medium additives are purified from animal sources other than serum, it is also important to ensure that microbial contamination with bacteria, viruses, fungi, parasites, or prions is excluded (Ong et al., 2021; Smith-Uchotski & Wanjiru, 2022; FAO & WHO, 2023; Gu et al., 2023).

Medium components can also be produced using microorganisms such as fungi or bacteria. Precision fermentation is a biotechnological process in which microorganisms are programmed with genetic information to produce proteins from another organism, such as animals or humans (Singh et al., 2022). For cultured meat production, precision fermentation can be used, for example, to generate medium components such as transferrin or insulin (Singh et al., 2022). While in principle this technique produces proteins that are identical in amino acid sequence to their animal or plant counterparts, small modifications to the proteins may occur due to the production process. These changes may potentially trigger allergic reactions (FAO & WHO, 2023). In addition, product purity may not be optimal, and microbial residues (such as endotoxins) or allergenic substances may remain in the final preparation (FAO & WHO, 2023). For such molecules to exert harmful effects on consumers, they must persist as part of the final cultured meat product. Below, an example is provided of a possible cultured meat additive produced through precision fermentation, and the associated potential hazard for allergenicity.

Cultured meat is generally paler in colour than conventional meat because the oxygen conditions under which cells are grown suppress the expression of the muscle protein myoglobin. Myoglobin is important for oxygen storage and transport and, due to its iron-binding capacity, provides the red colour characteristic of meat (Sakata & Honikel, 2001). To mimic the colour of conventional meat, cells could be cultured under hypoxic conditions or genetically engineered to express myoglobin at higher oxygen levels (Simsa et al., 2019). Alternatively, myoglobin or a comparable component could be added to the cultured meat product. One potential additive is soybean haemoglobin, a protein structurally related to myoglobin. Recombinant soybean haemoglobin can be synthesized by the yeast *Pichia pastoris*, which is commonly used for protein expression in cosmetics and pharmaceuticals (Zhang et al., 2020). During the production of soybean haemoglobin for use in meat alternatives, residual proteins from *Pichia pastoris* were detected (Jin et al., 2018). These proteins, however, showed no cross-reactivity with known allergens, suggesting that this protein additive does not pose a risk of food allergies (Jin et al., 2018). Nevertheless, the detection of residual *Pichia pastoris* proteins highlights the importance of evaluating the allergenicity of any cultured meat additive. Such evaluations should be conducted for each individual additive.

The application of scaffolds is unique to cultured meat production. Scaffolds enable the formation of three-dimensional muscle tissue. They can also provide cells with oxygen and nutrients, as well as (mechanical) signals that promote cell differentiation. Scaffolds may be composed of natural or synthetic materials, or combinations thereof. Natural materials can be derived from animal sources, such as collagen, fibrin, and gelatine, or from plant sources, such as alginate, cellulose, and soy protein (Campuzano & Pelling, 2019; Chen et al., 2022; Santos et al., 2023). Other scaffold materials include chitosan or mycelium, produced by fungi (Campuzano & Pelling, 2019; Chen et al., 2022; Santos et al., 2023). Scaffold materials may also be produced through precision fermentation. When animal-derived materials are used, microbial contamination may occur (Ong et al., 2021; Smith-Uchotski & Wanjiru, 2022; FAO & WHO, 2023; Gu et al., 2023). When plant-derived materials are used, allergens may potentially be present (Ong et al., 2021; Smith-Uchotski & Wanjiru, 2022; FAO & WHO, 2023; Gu et al., 2023). For materials derived from fungi or produced through precision fermentation, microbial residues may remain in the cultured meat product, posing a potential hazard to consumers.

In addition to medium additives and scaffolds, instruments and materials used during the cultivation phase, as well as human handling (such as cell thawing), can serve as sources of microbial contamination. If laboratory workers themselves are infected with a pathogenic microorganism and shed it, these microorganisms could be transmitted to the culture, for instance via airborne aerosols. Respiratory viruses are one example. Furthermore, microorganisms present in the laboratory environment can enter cultured cells through unhygienic handling practices.

Viral contamination may also occur through cross-contamination from other cell lines within the laboratory or facility. Contamination can likewise arise when instruments and materials are not adequately cleaned and disinfected. The bioreactor, as the primary cultivation environment, is

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particularly vulnerable to contamination with pathogenic microorganisms. Ensuring fully sterile conditions in bioreactors is complex. The inlet and outlet ports for gases or medium represent critical points of vulnerability, as these are difficult to clean and disinfect. Spore-forming bacteria (such as *Bacillus* species) can be particularly challenging to eliminate using disinfection methods. Bacterial spores can survive extreme conditions, such as heat treatment. Once germinated, they may proliferate, contaminating the culture and/or producing toxins that could persist in cultured meat. Fungal spores may also contaminate culture environments, including bioreactors and instruments. Spore-forming fungi (such as *Aspergillus* species) can likewise produce toxins (mycotoxins) that may end up in cultured meat and pose a health risk to consumers. Mycotoxins may also be introduced into cultured meat through contaminated culture medium components.

3.2.2 Chemical hazards

Chemical hazards during the proliferation and differentiation phase of the culture may originate from culture medium additives and scaffolds, or from other materials that come into contact with the culture, such as bioreactors and other equipment.

In vitro cultured cells require a variety of nutrients, growth factors, and hormones in order to proliferate and differentiate. These substances are supplied to the cells through the culture medium. The components of cell culture media can broadly be divided into two groups. The first group comprises the basal medium, a solution containing essential factors for cell growth, such as amino acids, vitamins, glucose, and inorganic salts (Zhang et al., 2020). The second group consists of specific factors added to stimulate cell proliferation and differentiation. Supplements that may be included are, for example, a derivative of L-ascorbic acid to stimulate collagen formation, transport proteins (such as transferrin and albumin), antioxidants (such as selenium), growth factors (such as FGF-2 (fibroblast growth factor), TGF-β1 (transforming growth factor beta), EGF (epidermal growth factor)), and hormones (such as insulin) (Chen et al., 2011; Kolkmann et al., 2022; Messmer et al., 2022; Stout et al., 2022).

These nutrients, growth factors, and other medium supplements are intended to support the growth and development of cultured cells into a final edible cultured meat product. If these additives are not fully metabolized during the cultivation process, residues may remain in the final product. Such residues may pose risks to consumers if concentrations exceed health-based threshold values. It is known that some growth factors and hormones are highly conserved between humans and animals; for example, bovine and human IGF-1 are structurally identical (Juskevich & Guyer, 1990). In such cases, interactions could occur between animal-derived molecules in the cultured meat product and human receptors in the consumer's body, potentially leading to undesirable biological effects. For this to occur, the molecules from the culture medium would need to be incorporated into the cultured meat and remain in sufficient amounts in the final product. Furthermore, these residues must withstand food processing and gastrointestinal passage. It is known that some growth factors are resistant to degradation by heat (Ewaschuk et al., 2011; Escuder-Vieco et al., 2018).

The consumption of animal-derived growth factors from the culture medium that remain in the final product may be undesirable. Intake of excessive hormones via food can lead to hormonal imbalances and adverse health effects. The use of exogenous hormones as growth promoters has therefore been banned in livestock systems for conventional meat production in the EU since 1981, due to potential human health risks from residues persisting in meat[11]. It is therefore essential to determine the levels of such medium-derived additives in the final cultured meat product. Even for components that occur naturally in conventional meat, it may be important to assess whether levels differ in cultured meat, and whether such deviations are safe for human consumption.

Although some reported medium additives may be approved for medical use, they are not necessarily safe for human consumption or authorized for this purpose. An example of a reported medium additive in cultured meat production is a p38 inhibitor. Inhibition of p38 signalling suppresses the loss of stem cell characteristics in bovine muscle stem cells, allowing them to continue dividing for longer (Ding et al., 2018). However, the p38 inhibitor is a chemical compound that is unsafe for human consumption.

In addition to growth stimulators, substances may also be added to the medium to promote cell differentiation. To produce muscle or fat cells, stem or progenitor cells must undergo

[11] Council Directive 81/602/EEC of 31 July 1981 concerning the prohibition of certain substances having a hormonal action and of any substances having a thyrostatic action

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differentiation steps. In some protocols, hormones and/or small molecules are used to induce differentiation. For instance, the differentiation of adipocytes often involves the use of dexamethasone, isobutylmethylxanthine (IBMX), insulin, and thiazolidinediones (Dufau et al., 2021; Kirsch et al., 2023). Although these compounds are used in the treatment of various medical conditions, they are not necessarily suitable for use in food products—for example, because of the acute oral toxicity of IBMX (Louis et al., 2023; Mitić et al., 2023).

For economic reasons, recycling of culture media may be an option for cultured meat producers (Ong et al., 2021). In such cases, it is essential to ensure that no toxic metabolites accumulate that could persist in the final cultured meat product.

Scaffolds can be made from synthetic materials, including various polyesters such as polyamide and polyethylene, which have previously been designated as safe for use in food preparation (Campuzano & Pelling, 2019; Chen et al., 2022; Santos et al., 2023). Other materials such as polyethylene glycol (PEG), polyglycolic acid (PGA), or polylactic acid (PLA) may also be used. These are biodegradable thermoplastic polymers. If scaffold materials are used that are not suitable for human consumption, the scaffold must be removed or degraded by the end of the culture process. Residues of microcarriers or scaffolds that are not fully degraded or removed and remain in the final cultured meat product may pose risks to consumers.

Scaffolds may undergo various physical or chemical processes (Oryan et al., 2018; Bomkamp et al., 2022). Transformation processes may occur, such as polymerization or surface modification. Scaffolds may also undergo degradation during cultured meat production.

An important aspect of scaffolds is the integration of so-called crosslinkers. Crosslinkers are used to assemble and stabilize the scaffold and to prevent disintegration. They may also be used to add substances as side groups to the scaffold. Various types of crosslinkers can be applied, including chemical crosslinkers or those activated by UV irradiation (photo-initiators). Above certain concentrations, chemical crosslinkers or photo-initiators can be toxic. If residues of these substances remain in the final cultured meat product, they may pose a risk to consumers. Transformation and degradation processes of scaffolds may also generate by-products that could give rise to new food safety risks (Ong et al., 2021; Smith-Uchotski & Wanjiru, 2022; FAO & WHO, 2023; Gu et al., 2023). For these reasons, residual concentrations of such by-products in the final edible product should be determined.

In addition to medium additives and scaffolds, substances and materials that come into contact with the cell culture may be a source of chemical contaminants that can end up in the final cultured meat product. Possible sources of contaminants include the culture container (flask or bioreactor) and the instruments used during cultivation. A potential hazard is the leaching of (heavy) metals from the bioreactor casing or the instruments.

Many plastic products are used during cell culture, such as culture flasks or dishes. Microplastics are small particles resulting from the degradation of plastic materials and are frequently detected in food (Wen et al., 2024). At present, the toxicity of microplastics in food is unknown (Ong et al., 2021; Smith-Uchotski & Wanjiru, 2022; FAO & WHO, 2023). Microplastics could also alter the properties of other food components through interactions with them. Plastics used for cell culture could be a source of endocrine-disrupting chemicals, such as bisphenol A (BPA) and dibutyl phthalate (DBP) (Manikkam et al., 2013; Benny et al., 2022). These substances could potentially accumulate in in vitro cultured cells. It has previously been shown that endocrine-disrupting chemicals can migrate from plastic packaging into food (López-Cervantes & Paseiro-Losada, 2003). The same phenomenon has been observed for cells cultured in plastic materials for in vitro fertilization (Ishii et al., 2013).

Bioreactors and other reusable culture instruments must be cleaned and disinfected between successive cultures to prevent the transfer of pathogens or contaminants. Residues of cleaning and disinfecting agents that remain in bioreactors may potentially end up in cultured meat and pose a risk to consumers. For chemicals used to clean equipment and materials, it is important to verify that no toxic residues remain in the final product (Ong et al., 2021; Smith-Uchotski & Wanjiru, 2022; FAO & WHO, 2023).

3.3 Hazards associated with cell harvesting and further food processing

In the final stage of cultured meat production, the cell mass is harvested and further processed into edible cultured meat products. Detailed information on food processing methods for cultured meat is largely not publicly available. Therefore, assessing potential microbiological and chemical hazards is only possible to a limited extent. Below, some potential microbiological and chemical hazards of the food processing stage are described.

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3.3.1 Microbiological hazards

During cultured meat production, microbial contamination can occur during cell harvesting and subsequent food processing, packaging, and preparation. This includes environmental microorganisms that are not strictly animal-associated but can survive in food processing facilities and on equipment, such as *Salmonella* spp., *Escherichia coli*, *Listeria monocytogenes*, and *Staphylococcus aureus* (Habowski & Sant'Ana, 2024; Powell et al., 2025). Contamination during processing can occur due to unhygienic handling or cross-contamination from insufficiently disinfected instruments or materials. Contaminated reagents or infected personnel are also possible sources of contamination. Epidemiological data from the conventional meat industry show that infected slaughterhouse workers are frequently involved in the transmission of foodborne viruses, such as hepatitis A and norovirus, via raw meat (Habowski & Sant'Ana, 2024; Powell et al., 2025). Cultured meat could be similarly contaminated with viral pathogens from infected staff during the processing stage if hygiene protocols are not properly followed. Because the cells die during or shortly after harvest, viruses cannot replicate further. However, already formed viral particles may survive in chilled or frozen products.

Although muscle tissue from healthy animals is sterile, it inevitably becomes exposed to microorganisms during slaughter and subsequent food processing, leading to the development of a microbiome on conventional meat products. Similarly, a microbiome will form on cultured meat during food processing (Habowski & Sant'Ana, 2024; Powell et al., 2025). The composition of this microbiome and the influence of present microorganisms on the quality and safety of cultured meat during processing, transport, chilling, and storage are currently unknown. Likewise, the potential health effects on consumers are unclear.

There are many similarities in processes and potential microbiological hazards between conventional meat and cultured meat production, particularly during the food processing stage. Cooking cultured meat for consumption, as is done with conventional meat, is a critical control step for microbiological hazards. Many viral and bacterial pathogens are inactivated by sufficient heat during meat preparation. Additionally, storage conditions of the product prior to preparation are important for controlling the growth of bacterial pathogens present on the meat. Developing preparation and hygiene guidelines for cultured meat, similar to those for conventional meat, is essential for managing microbiological food safety hazards. Existing quality management and risk control systems focused on food safety, such as HACCP (Hazard Analysis and Critical Control Points) and GMP (Good Manufacturing Practices), as well as frameworks for safe, reliable, and reproducible cell culture procedures, such as GCCP (Good Cell Culture Practices), provide a basis for identifying, monitoring, and managing potential risks. Adapting these methodologies from conventional meat production and pharmaceutical processes to the specific characteristics of cell-based cultured meat production can significantly contribute to controlling microbiological hazards.

3.3.2 Chemical hazards

Food processing and preparation can unintentionally generate chemical substances that end up in food products. These substances are referred to as process contaminants. Examples of process contaminants include acrylamide, polycyclic aromatic hydrocarbons (PAHs), furans, 3-MCPD esters (3-monochloropropanediol), and glycidyl fatty acid esters. At present, it is unknown whether and to what extent the formation of these substances during the processing and preparation of cultured meat differs from conventional meat. It is also unclear how components of cultured meat respond to food processing steps, such as heating, or interactions with other ingredients. Further scientific research is needed to determine whether substances that are hazardous to consumers are formed during the processing phase. If scaffolds are part of the final product, it is important to assess whether food processing steps, such as heating, could lead to the formation of harmful substances derived from the scaffold.

4 Hazard characterisation

Hazard characterization constitutes the second step in the risk assessment procedure according to the Codex Alimentarius. During hazard characterization, the effects of the identified hazards on human health are assessed. The hazards described above have only been identified on a theoretical basis as potentially associated with the consumption of cultured meat. No data are available regarding the actual presence of microbiological agents or chemical substances in cultured meat. For this reason, BuRO has not conducted further characterization of the identified hazards. For characterization of certain hazards that may be comparable to conventional meat—such as the transmission of zoonotic pathogens through meat consumption or the transfer of

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antimicrobial resistance via meat consumption—reference is made to the risk assessment of the red meat supply chain[12].

5 Exposure assessment and Risk characterisation

The CoP prescribes that participants in Dutch tasting events may taste cultured meat a maximum of 10 times per year. No information is provided regarding portion sizes. Furthermore, no data are available on the presence of pathogenic microorganisms or chemical substances in cultured meat offered at Dutch tasting events. As a result, the exposure of participants to microbiological or chemical hazards cannot be determined. Due to the short-term intake, hazards with acute health effects are particularly relevant for this risk assessment. Non-acute effects after short-term intake could also occur, for example in the case of a long incubation period. A possible health effect from short-term exposure is the occurrence of an allergic reaction. No data are available on the potential presence of allergens in cultured meat. The CoP requires that participants declare they have no known allergies.

Participants could become acutely ill if pathogenic microorganisms are present in cultured meat. Some pathogenic microorganisms can cause infection following short-term intake but have incubation periods of several weeks to months. During cell culture, maintaining sterility is crucial, as bacterial contamination could cause the culture to fail. This may indicate that the bacterial contamination level in cultured meat products is low. On the other hand, laboratory conditions during culture could favour the survival of microorganisms not commonly found in conventional meat production chains.

For infection to occur in a consumer, pathogenic microorganisms in contaminated cultured meat must be present, survive processing and preparation, be capable of zoonotic (animal-to-human) or human-to-human transmission, and be orally transmissible. No scientific analyses are available on the specific microorganisms present in cultured meat. Due to the absence of data on the type and quantity of microorganisms in cultured meat, no further exposure assessment or risk characterization regarding microbiological agents can be conducted.

For many chemical substances, effects occur only after prolonged or chronic exposure, with consequences observable only after many years. In the case of cultured meat tastings, exposure is short-term (acute) rather than chronic. The intake amount is calculated as the product of the measured or estimated concentration of the substance in cultured meat and the portion size. Risk characterization can only be performed when the intake amount can be compared with the health-based guidance value for the substance or substance group. For cultured meat tastings, this would involve a health-based guidance value for short-term exposure, i.e., an acute reference dose (ARfD). However, no literature is available reporting concentration data for chemical substances in cultured meat. Due to the absence of these concentrations, no exposure assessment or risk characterization can be performed.

6 Risk assessment of novel foods and tastings in the EU and abroad

Before a novel food can be authorized for the European market, EFSA carries out a risk assessment regarding the food safety of the product. Under the Dutch CoP procedure, cultured meat producers wanting to organize tastings are required to submit an application, which is assessed by a dedicated expert committee established for this purpose. Below, the food safety aspects assessed by EFSA and the Dutch expert committee are described, as well as the type of information food producers are expected to provide to enable these risk assessments. Subsequently, the procedures for market authorization of cultured meat and/or other novel foods, and for organizing tastings in Singapore and the United States, are outlined.

6.1 European Union

EFSA has developed a document containing scientific and technical guidance for submitting a dossier for the authorization of a novel food (EFSA NDA Panel et al., 2021). This document is not specifically tailored to cultured meat. It provides an overview of the data EFSA requires to assess whether a novel food poses no risks to human health. The required information includes a detailed description of the novel food and an exact description of the production process. The composition

[12] Advies van BuRO over de risicobeoordeling roodvlees- en grofwildketen | Advies | NVWA

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of the product must be specified, along with the chemical properties of its various components, as well as nutritional value and microbiological parameters that characterize the food. Any impurities, by-products, chemical or microbiological contaminants, and residues must also be identified. In addition, absorption, distribution, metabolism, and excretion (ADME), as well as potential toxicological effects and allergenicity, must be analysed and data must be submitted. Finally, the proposed use and expected intake levels of the novel food must be reported. It is the applicant's responsibility to demonstrate, based on the data provided, that the novel food is safe for consumption. If potential hazards are identified, their relevance must be described in relation to the expected intake of the food.

6.2 Dutch tastings

The CoP for cultured meat tastings in the Netherlands claims to broadly follow EFSA's guidance for novel foods, but in a simplified form. Whereas EFSA evaluates novel foods in the context of long-term or chronic consumption, tastings involve short-term intake (a maximum of 10 times per year per participant). According to the CoP, the documentation required for a tasting application in the Netherlands must include: a description of the product, including the cells and the source animal, and an outline of the production process. In addition, potential allergenicity must be addressed, as well as information on whether genetic modification is applied, and on the microbiological status of the final product. The main focus of a tasting application must be an overview of the cultured meat composition. For each component, the measured or calculated concentration in the cultured meat product must be determined, and the total intake of the substance calculated. This intake is then compared with an established safety threshold, such as the Acceptable Daily Intake (ADI) or the Acute Reference Dose (ARfD), while taking into account that tastings involve short-term exposure.

6.3 Singapore

Singapore became the first country in the world to formally approve a cultured meat product for market entry in December 2020. Like the European Union, Singapore has a regulatory framework for novel foods and considers cultured meat to fall under this category. According to the Singapore Food Agency (SFA), novel foods are ingredients and products that lack a "history of safe use." A history of safe use means that the food has been consumed as part of the diet of a significant proportion of the population for at least 20 years without adverse health effects. Any food without such a history must receive approval from the SFA before it can be placed on the market.

The SFA has issued guidelines for submitting an application for market authorization of a novel food. These guidelines closely resemble EFSA's framework for novel foods in the EU. However, the SFA guidelines include a dedicated section specifying the information requirements for cultured meat. An application must include a description of the cultured meat product and its composition. In addition, background information on the cells and their origin (cell lines or stem cells), as well as a description of the methods used for cell selection and isolation must be included. Also, a full list of chemical substances added to the cells during production, including all medium additives, growth factors, veterinary drugs, and scaffold materials must be provided. For all components not known to be suitable for human consumption, and for any unintentionally formed metabolites, a risk assessment must be carried out. In addition, biopsy material must be free of animal diseases, and culture media and cell lines must remain uncontaminated with infectious agents (such as viruses, bacteria, fungi, and prions) throughout the production process. Any genetic modification and genetic instability of the cells must also be reported, along with an assessment of whether these could result in the production of substances that may pose food safety risks.

The SFA guidelines also outline procedural requirements for organising tastings of novel foods. Novel foods that have not yet undergone a full safety assessment may not be offered for consumption at promotional events for the product or manufacturer, but they may be used for product development purposes. For this, an exemption must be requested from the SFA. Such an exemption application must include a description of the product, its source, and the production process. In addition, a risk assessment for acute hazards must be conducted, addressing microbiological agents, contaminants, toxins, and allergens that could potentially be present in the product to be tasted. It is the responsibility of the producer to demonstrate that short-term intake of the novel food does not result in adverse health effects and that the tasting can be conducted safely. Participation in tastings is limited to a maximum of 30 preselected volunteers, and tastings are not open to the general public. Any adverse health effects must be reported for up to two weeks following the tasting event.

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6.4 United States of America

The United States of America (USA) does not have a formal definition of novel foods and therefore has no dedicated regulatory framework for them. Food ingredients and additives must be approved by the Food and Drug Administration (FDA) before they can be marketed, unless they can be designated as "generally recognized as safe" (GRAS). With GRAS status, a food product can be marketed without going through the FDA's approval process.

GRAS status is based either on significant use prior to 1958 or on scientific evidence that the substance or product is safe. It is the producer's responsibility to demonstrate product safety. The producer may provide existing scientific data or assemble a panel of qualified food safety experts to confirm that sufficient scientific evidence exists showing that the food can be safely consumed as intended. A producer notifies the FDA of the GRAS status of an ingredient or product. The FDA may then conclude that there is insufficient information to support GRAS status, or it may state that it has no further questions regarding the producer's safety conclusion. This does not constitute explicit approval but indicates that the FDA does not dispute the producer's determination.

In a 2019 agreement, the FDA and the United States Department of Agriculture (USDA) Food Safety and Inspection Service (FSIS) defined their respective roles in overseeing food made from in vitro–cultured animal cells.[13] For foods made from cultured animal cells of fish (other than catfish), seafood, and exotic game species, the FDA oversees the entire production process. For foods derived from cultured animal cells of livestock, poultry, and catfish, the FDA and USDA-FSIS share oversight. The FDA oversees the production process up until the point at which the cells are harvested from the bioreactor. This includes tissue collection, cell isolation, cell line development and storage in cell banks, as well as all ingredients and components used in the culture process. The FDA conducts a consultation in which the culture process is evaluated for food safety of the final product. This consultation process is not an approval procedure. A completed consultation means that, after a thorough evaluation of the information provided by the producer, the FDA has no further questions regarding the producer's conclusion about food safety. The USDA-FSIS takes over oversight at the stage of cell harvest and for subsequent food processing steps after harvest, including packaging and labelling. Any meat producer, including cultured meat producers, must undergo a USDA-FSIS inspection to ensure compliance with statutory food safety requirements before products can be marketed.

Commercial tastings are not permitted in the United States, and novel foods may not be offered to the general public, media, or potential investors. Tastings for research purposes are not formally prohibited. Only company employees or contracted test subjects may participate in such tastings. It is the producer's responsibility to demonstrate that the product offered for tasting is safe for human consumption.

6.5 Comparison between countries

Table 2 provides an overview of several aspects of legislation and regulations concerning novel foods, cultured meat, and tastings in different countries and in the EU as a supranational entity.

Both the EU and Singapore have legislation on novel foods and requirements for the information that producers must provide in a safety dossier when submitting an application for market authorization of a novel food. The EU and SFA guidelines show many similarities. Both sets of guidelines stipulate that all information relevant to the food safety of the novel food must be submitted. A brief summary of these information requirements has been outlined in the descriptions above for each respective jurisdiction. Unlike the EU guidelines, Singapore's guidelines include provisions specifically tailored to cultured meat as well as targeted requirements for organizing tastings of novel foods. The information requirements for an application to organize novel food tastings in Singapore and cultured meat tastings in the Netherlands show many similarities. Whereas applications for market authorization in the EU and Singapore are focused on long-term or chronic consumption, tastings involve short-term intake and therefore require a risk assessment for acute hazards. The requirements for applications for market authorization of novel foods in the EU and Singapore, and for organizing (cultured meat) tastings in Singapore and the Netherlands, are formulated in such general terms that they provide insufficient insight into what food safety information is actually submitted by food producers. Few specific potentially hazardous chemical substances or microbiological contaminants are explicitly mentioned in the requirements.

[13] Formal agreement between the U.S. department of health and human services food and drug administration and U.S. department of agriculture office of food safety

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Table 2 Comparison of novel food procedures across different countries.

Procedure	European Union (EU)	Singapore	United States of America (USA)	Dutch tastings
Legislation on novel foods	Yes	Yes	No	N/A
Guidelines for safety dossier novel foods	Yes	Yes	No	N/A
Guidelines specific for cultured meat	No	Yes	No	Yes, CoP procedure
Guidelines specific for tastings	No	Yes	No	Yes, CoP-procedure
Safety dossier assessed by	EFSA	SFA	FDA and USDA-FSIS	Committee established specifically for this purpose

N/A = Not applicable. The column 'Dutch tastings' specifically concerns the procedure established to enable Dutch tastings of cultured meat, outside the existing European legislation applicable in the Netherlands that relates to novel foods.

The US does not have formal legislation regarding novel foods and therefore no guidelines for submitting a safety dossier on novel foods, nor for organizing tastings. However, the US does follow an evaluation procedure when a food product has not yet been marketed domestically. In such cases, a dossier is submitted for consultation, which is evaluated until there are no further questions about the producer's conclusion regarding food safety. The risk assessments and/or evaluations conducted for market authorization of novel foods in the EU, Singapore, and the US, are conducted by food safety authorities. In contrast, the assessment of applications for Dutch tastings is performed by a committee established specifically for this purpose, which is not affiliated with a recognized food safety authority.

7 Analysis of uncertainty

BuRO has focused on identifying the chemical and (micro)biological agents that may be present in a cultured meat product and that could potentially have harmful health effects for humans. For this purpose, BuRO conducted a review of the scientific literature. In the scientific literature, descriptions are primarily available of the cultured meat production process. Based on these descriptions, potential hazards associated with the consumption of cultured meat have been inferred. These hazards are purely theoretical, as there are few to no measurement data available on the actual presence of pathogenic microorganisms and chemical substances in cultured meat. In addition, data are lacking on the specific composition of cultured meat products offered by Dutch producers during tastings. The dossiers submitted when applying for permission to organize tastings are not publicly accessible and were therefore not reviewed by BuRO. Information on the portion sizes consumed during tastings is also missing. Due to the limited availability of information, there is a high degree of uncertainty regarding the food safety risks of organizing cultured meat tastings.

8 Conclusion

Cultured meat is considered a novel food in Europe and falls under the legislation laid down in the Novel Food Regulation (EU) 2015/2283. In order to place a novel food on the market in Europe, a system has been established to safeguard food safety. This system includes conducting a risk assessment by EFSA to determine whether consumption of the novel food poses risks to human health. A positive opinion from EFSA forms the basis for market authorisation in the EU, followed by approval by the European Commission and the EU Member States. The European legislation for novel foods has been incorporated into Dutch law and is enforced in the Netherlands by the NVWA. The NVWA's supervision is based on the proven food safety system as established in European and Dutch legislation. In the Netherlands, however, a procedure has currently been set up to enable cultured meat tastings, which falls outside the framework of the established food safety system. Food safety during tastings is assessed by an expert committee set up specifically for this purpose. This committee is not affiliated with a recognised food safety authority, and little is known about its composition.

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Guidelines have been established for EFSA's risk assessment of novel foods for market authorization, specifying the information that producers must provide about the food. The analysis of procedures in other countries where cultured meat has been authorised for the market shows that the guidelines prescribing which information must be provided by producers in applications for novel food market authorisation display many similarities. These state that all data relevant to food safety must be supplied. However, few specific potentially hazardous chemical substances or microbial contaminations are mentioned. The risk assessments and/or evaluations carried out for market authorisation of novel foods in the EU, Singapore and the USA, are conducted by food safety authorities.

Under the Dutch CoP procedure for organising tastings, the evaluating committee must be provided with information on the composition of the cultured meat product and on the safety of acute intake. The requirements for submitting an application to organise cultured meat tastings and the associated safety dossier are formulated in such general terms that they do not provide sufficient insight into which food safety information is actually supplied by cultured meat producers. The information provided is confidential and not publicly accessible, and therefore also unknown to the supervising authority. It is undesirable and inefficient for the supervisor to have no insight into the safety assessment of products that fall under its supervision. The assessment procedure of food safety during cultured meat tastings is non-transparent for the supervisor, and the reproducibility of the process is uncertain.

Based on insights into the production process, BuRO has identified hazards that may potentially be associated with consumption of cultured meat. The production process of cultured meat can be divided into the collection of starter cells, biomass production through cell proliferation and differentiation, and cell harvesting and subsequent food processing. During these various stages, (micro)biological and chemical hazards may arise that could pose potential risks to the consumer of cultured meat. The exact production process, the origin of the cells, the ingredients of the culture medium, or the nature of the structural components of cultured meat intended for Dutch tastings are unknown.

Microbiological agents such as bacteria, viruses, fungi, parasites, and prions can cause disease when ingested through food. The type of microorganism and the timing of contamination largely determine the outcome of a microbial contamination of cultured meat. If the culture is not disrupted and a finished cultured meat product is formed that is contaminated with pathogenic microorganisms, then consumption of cultured meat may cause illness. Culturing cells under laboratory conditions may potentially promote the growth and survival of microorganisms other than those known from conventional meat production chains.

A possible source of contamination is microorganisms present in the biopsy tissue taken from the source animal. Visual inspection of the animal and the harvested tissue, as well as information on the animal's health status, can provide insights into potentially present pathogenic microorganisms. However, it is also possible that microorganisms pathogenic to humans are present in the animal without visible symptoms. Other potential sources of microbial contamination include culture medium additives or structural components derived from an animal source, or produced using precision fermentation by microorganisms. Contamination may also occur through the use of contaminated culture materials and instruments, from the culture environment, or from the personnel performing the procedures. Another possible hazard is the formation of toxic or allergenic proteins due to spontaneous or induced genetic instability of the cultured cells.

Chemical hazards are (residues of) chemical substances that may have toxic effects on a consumer of cultured meat. For these substances, they must remain in sufficiently high concentrations in the final cultured meat product to cause adverse health effects. One source of chemical contamination of cultured meat may be residual chemical substances in the biopsy tissue, including environmental contaminants, residues of veterinary medicines and feed additives, and natural toxins. Another potential source includes bioactive molecules, such as hormones and growth and differentiation factors, that are added through the culture medium and may remain in cultured meat. In addition, breakdown or transformation by-products of structural components of cultured meat may have toxic effects. Finally, materials that come into contact with the culture, such as the culture container and instruments used during culturing, may be sources of chemical contamination. Possible hazards include heavy metals from the bioreactor, microplastics, and residues of cleaning and disinfecting agents.

In the case of tastings, consumption of cultured meat is limited to short-term intake, meaning that only hazards with acute health effects are relevant. For many chemical substances, effects occur only after long-term or chronic exposure. Microbiological contamination of cultured meat may

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cause acute infection in tasting participants. When cultured meat enters the market and becomes part of the regular diet of the Dutch population, (micro)biological and chemical agents with adverse health effects from long-term intake will also be of concern. The risks associated with long-term or chronic intake are unknown. Data on the actual presence of pathogenic microorganisms and concentrations of chemical substances in cultured meat are lacking. BuRO also has no access to the confidential contents of the applications for the tastings that have already taken place in April and July 2024. Consequently, the (micro)biological and chemical health risks for participants in Dutch cultured meat tastings cannot be determined. It therefore cannot be ruled out that risks are associated with consuming cultured meat during tastings held in the Netherlands. Due to a lack of transparency in the assessment process, it is unclear whether potential food safety risks are adequately addressed and evaluated in a sound and reproducible manner under the current procedure in place.

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10 Appendix

Appendix 1. Literature review method

Literature was searched in two databases on 15 January 2024: Scopus and Web of Science. The search was conducted using the English terms *'hazard'*, *'risk'* and *'safety'*, in combination with one of the following English terms for cultured meat: *Cultured meat, Cultivated meat, Artificial meat, Cell-based meat, Cell-cultured meat, Cell-cultivated meat, Cellular meat, Animal-free meat, In vitro meat, Lab-grown meat, Synthetic meat, Fake meat, Clean meat.*

Articles published before 2015 were excluded. The search results were sorted by relevance, after which the top 250 were assessed for usability based on the title, keywords and/or abstract. See Table 3 for an overview of the results of this search strategy.

Table 3 Search strategy and results.

Query 1	Query 2	Database	Number of results
Cultured meat OR Cultivated meat OR Artificial meat OR Cell-based meat OR Cell-cultured meat OR Cell-cultivated meat OR Cellular meat OR Animal-free meat OR In vitro meat OR Lab-grown meat OR Synthetic meat OR Fake meat OR Clean meat	AND Hazard	Scopus	319
		Web of Science	3917
	AND Risk	Scopus	2100
		Web of Science	3981
	AND Safety	Scopus	2142
		Web of Science	3989

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